

backend/rag/embeddings.py Documentation (Roman Urdu)

Yeh script Natural Language Processing (NLP) ki "Alfaz ko Numbers mn Tabdeel krne waali (Embedding)" motor hb!

File Ka Maqsad (Purpose)

Computer English nahi smjta bs math smjhta hy. Yeh file English context ko Array of Floats (e.g. `[0.12, 0.44, 0.9]`) ma encode krr ky dti hy ta k computer Vector distance/similarity check karr sky.

Code Blocks aur Functions ki Tafseel

1. `_load_model()`

- **Kya karta hai?:** Yeh offline HuggingFace system model `all-MiniLM-L6-v2` down/load krrtein hy lazy tariqay ky ssth. Is model k dimension (size) 384 hs.

2. `class EmbeddingEngine`

- **Tafseel:** Asal logic. Jab bb query athi e (`embed_query`), ya document parse hyotae (`embed_documents`), ye transformer ussey matrix array return kar deataa e!

3. Fallback Mode `_hash_embed()`

- **Zabaradast Trick:** Agr kici wajah ss User k pc men Python packages pz (e.g. `sentence-transformers`) masla krtk ya isntal hony sy fail rehty a, tou pura software CRASH naii hota! Yeh file apna "MD5 Hash technique" chalz kr fake embeddings bana lety ab jo guzara laik bilkl sahy kam chhlzatrty hb.

4. `cosine_similarity()`

- **Kya karta hai?:** Do mukhtlif vectors / jumlon ko aapas ma Match karny ki accuracy score float mb nikaltti hv.